

CLARION, IA, USA

WMO: 725458

Lat: 42.742N

Lon: 93.759W

Elev: 1162

StdP: 14.09

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04907

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.1	-4.5	-17.3	2.2	-9.2	-12.5	2.9	-5.6	33.1	19.4	30.8	16.4	7.9	320	0.601

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	90.9	75.2	89.5	74.5	86.1	73.0	79.6	86.9	77.7	84.9	75.5	83.4	13.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.3	148.3	84.5	75.1	137.6	82.7	72.9	127.6	80.2	44.0	87.0	42.1	84.9	39.9	83.5	88.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.7	25.8	23.2	-18.6	96.3	4.7	3.4	-22.0	98.8	-24.8	100.8	-27.4	102.7	-30.8	105.2		
(5)			-19.2	83.3	5.4	2.3	-23.0	84.9	-26.2	86.3	-29.2	87.6	-33.1	89.3	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.8	18.6	22.2	35.5	48.6	60.4	70.3	73.4	70.8	63.8	50.3	35.9	22.7
	DBStd	21.89	12.47	13.03	12.66	10.18	9.11	6.24	5.81	5.74	8.69	9.78	10.79	12.30
	HDD50	3800	974	780	476	144	18	0	0	0	6	117	436	849
	HDD65	7073	1439	1200	917	497	201	22	6	14	125	467	872	1313
	CDD50	3010	0	0	26	103	339	609	724	645	420	126	15	3
	CDD65	806	0	0	1	6	57	181	265	194	90	11	0	1
	CDH74	7726	0	0	17	122	625	1681	2554	1681	901	120	4	21
	CDH80	2602	0	0	2	30	204	577	935	522	291	29	0	12
Wind	WSAvg	10.4	11.6	11.7	12.2	13.2	12.6	9.9	7.1	6.3	8.2	10.3	11.4	11.0
Precipitation	PrecAvg	35.7	0.9	1.0	2.0	3.7	4.9	5.9	4.9	4.4	2.8	2.2	1.8	1.1
	PrecMax	48.3	1.7	1.7	4.5	7.2	9.4	10.6	10.3	12.2	5.7	6.7	4.4	2.5
	PrecMin	26.1	0.2	0.2	0.1	0.9	1.9	2.0	2.3	1.0	1.2	0.4	0.0	0.1
	PrecStd	5.8	0.4	0.5	1.2	1.7	1.9	2.4	2.2	3.0	1.2	1.5	1.2	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.8	56.9	75.4	83.8	90.8	93.2	95.0	91.3	91.2	84.4	71.9	60.6
		MCWB	42.4	52.1	60.2	61.5	67.7	74.6	79.4	76.2	73.2	66.7	58.1	54.9
	2%	DB	40.7	49.0	66.2	77.1	85.7	90.2	90.9	89.6	87.7	77.1	63.2	47.6
		MCWB	37.7	45.8	56.3	58.8	65.2	72.4	78.0	76.5	72.6	62.0	53.4	43.9
	5%	DB	36.6	42.8	60.6	71.9	81.1	86.4	88.3	85.8	82.3	72.2	56.7	41.2
		MCWB	34.8	39.2	53.3	56.0	63.6	71.1	76.5	75.6	68.5	60.6	48.8	37.7
	10%	DB	34.3	37.2	53.8	66.0	76.7	82.5	85.7	82.0	79.4	66.3	51.8	36.9
		MCWB	32.9	35.4	48.1	53.9	61.8	69.1	75.4	72.8	66.1	57.2	45.8	34.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.6	52.4	63.0	65.2	74.6	79.7	83.7	81.3	77.1	71.3	59.7	50.3
		MCDB	47.5	56.4	73.9	76.9	82.6	87.3	91.8	87.9	85.4	79.0	68.3	54.6
	2%	WB	37.8	45.6	57.7	61.5	70.3	77.1	80.3	78.9	74.4	66.4	54.3	42.5
		MCDB	39.9	48.0	63.7	72.5	79.2	85.3	87.4	86.3	83.1	73.0	61.7	46.0
	5%	WB	35.2	39.3	53.2	58.7	67.4	74.5	78.3	76.7	72.1	62.3	49.8	37.8
		MCDB	36.5	42.0	61.1	68.9	76.2	82.8	85.4	83.5	80.0	68.3	55.4	40.7
	10%	WB	32.8	36.0	47.4	55.2	64.5	72.1	76.2	74.4	69.6	58.5	45.6	34.8
		MCDB	34.1	37.7	52.6	63.8	72.3	80.2	83.9	80.9	75.8	65.6	50.3	37.0
Mean Daily Temperature Range	5% DB	MDBR	15.7	16.2	17.2	21.6	21.4	20.2	20.4	20.4	23.9	21.6	18.4	15.3
		MCDBR	17.4	18.8	27.5	31.8	27.4	23.9	22.3	22.4	26.6	28.2	27.2	20.2
		MCWBR	15.2	15.6	17.6	16.8	12.5	11.7	12.9	13.5	13.3	15.6	17.3	16.0
	5% WB	MCDWBR	16.9	18.2	25.8	26.1	21.9	19.6	19.9	19.4	22.2	22.5	24.4	18.2
MCWBR		15.0	15.6	17.5	16.1	12.8	12.7	13.6	13.5	12.7	15.3	17.0	15.2	
Clear-Sky Solar Irradiance	taub	0.279	0.291	0.317	0.370	0.398	0.422	0.428	0.412	0.384	0.338	0.296	0.274	
	taud	2.415	2.391	2.427	2.317	2.304	2.284	2.239	2.284	2.325	2.446	2.503	2.454	
	Ebn at Noon	275	290	295	285	278	271	267	268	267	269	268	268	
	Edh at Noon	24	29	32	39	41	42	43	40	36	28	22	21	
All-Sky Solar Radiation	RadAvg	588	915	1199	1496	1766	1961	2024	1723	1405	934	627	468	
	RadStd	63	78	120	139	130	158	110	118	96	120	49	36	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page